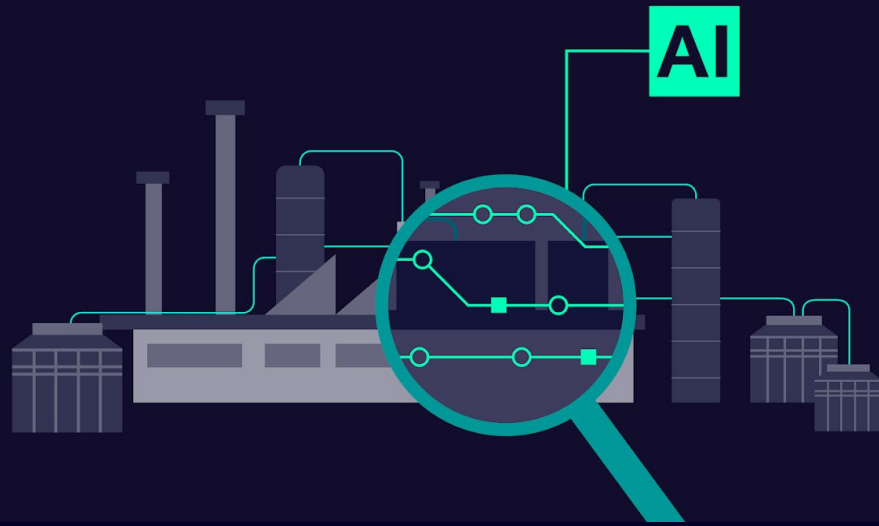




Digitalization, Data Analytics & AI in Gas Analytics

Oindrilla Chatterjee | Manager, AI and Digitalization, Adage

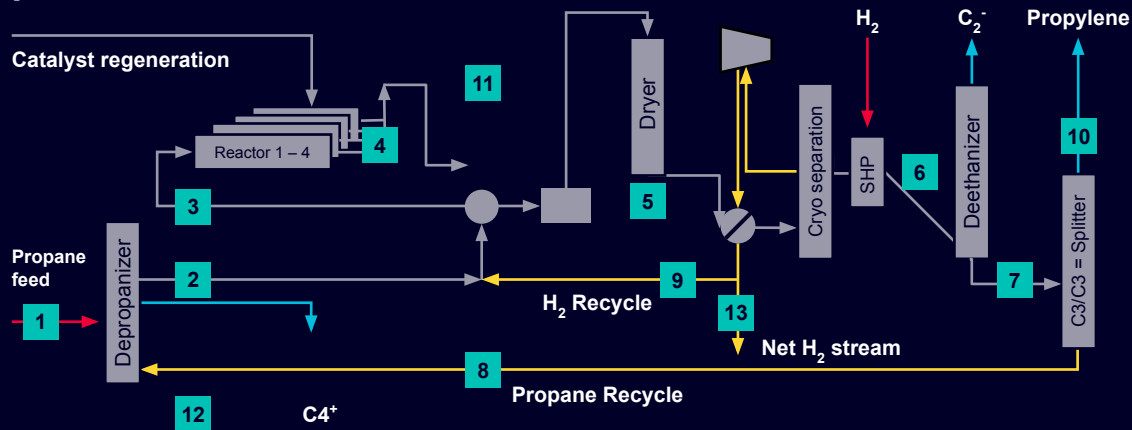
Background on ADAGE

- ADAGE is an Indian Gas Analytical Company with Global Presence.
- India's largest gas analytical operations with operations in India, KSA, UAE, USA
- Over 2 decades of experience in gas analyzers, Gas Chromatographs, CEMS, tunable diode analyzers



Process Analyzers are used for a wide range of applications and are crucial for any process

Example process plant with analytical measurement points



What if ...

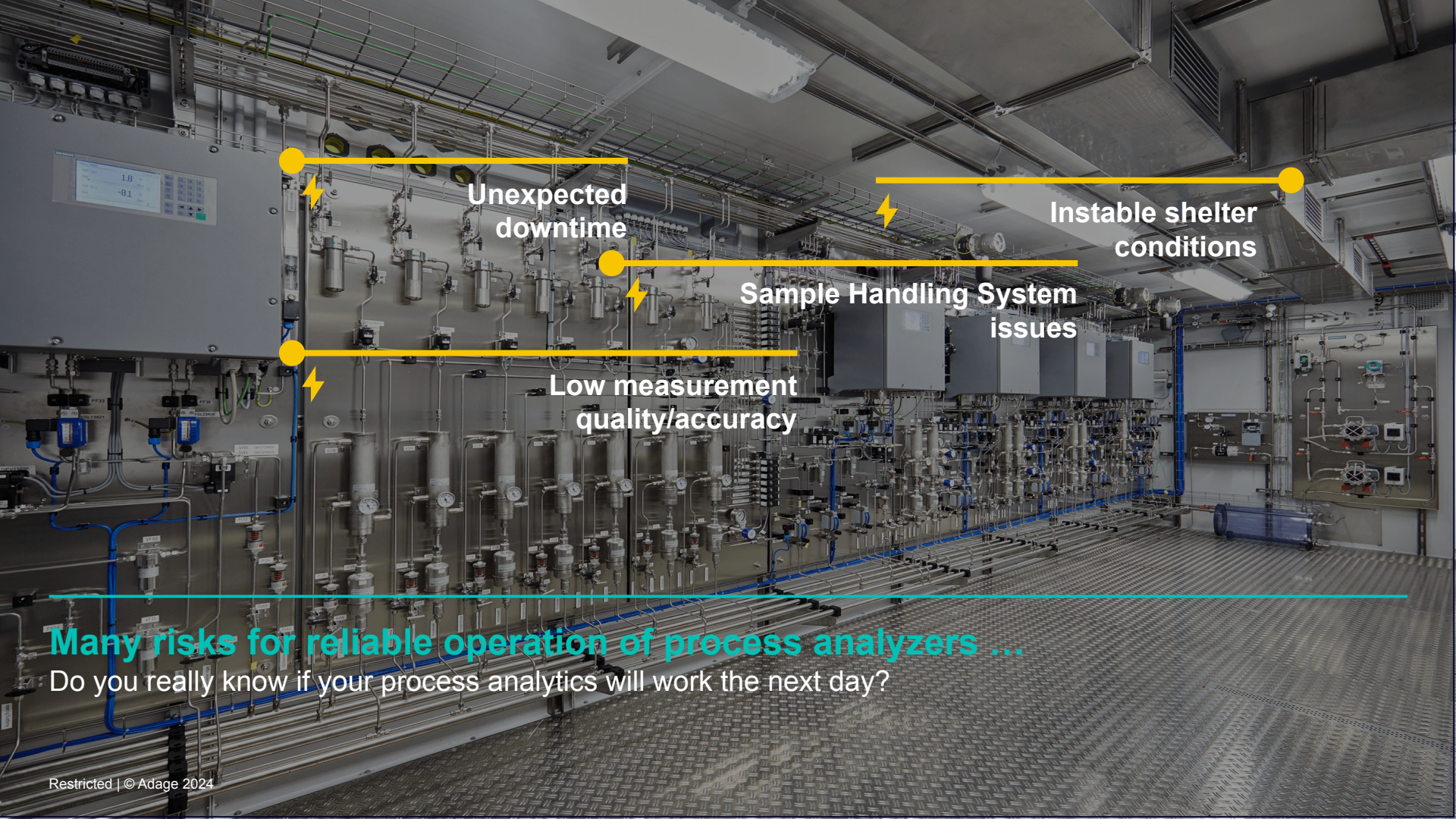
process analyzers don't deliver the correct information or fail?

Effects of an unreliable analyzer

- 01 Processes may be running in a sub-optimal state:**
 - Process yield may be lower than expected
 - Manufactured product may be of low quality
- 02 Operation costs may be higher than expected**
 - High maintenance costs
 - Excessive energy consumption
- 03 Violations of environmental or safety regulations**



ADAGE



Unexpected
downtime

Instable shelter
conditions

Sample Handling System
issues

Low measurement
quality/accuracy

Many risks for reliable operation of process analyzers ...

Do you really know if your process analytics will work the next day?



How can we master the challenges of Process Analyzer Technology and provide the **full value for optimized process control?**



Digitalization, Data Analytics & AI in Gas Analytics as (one) answer!

WHY?

Reliability



Optimization

Processes are in optimal state



Improvement of product quality & output

Manufactured product will have appropriate quality (not too good, not too bad, but exactly as required)



Lower OPEX

- Lower plant operation costs
- Less maintenance
- Less energy consumption



Increased yield

Increased plant production yield

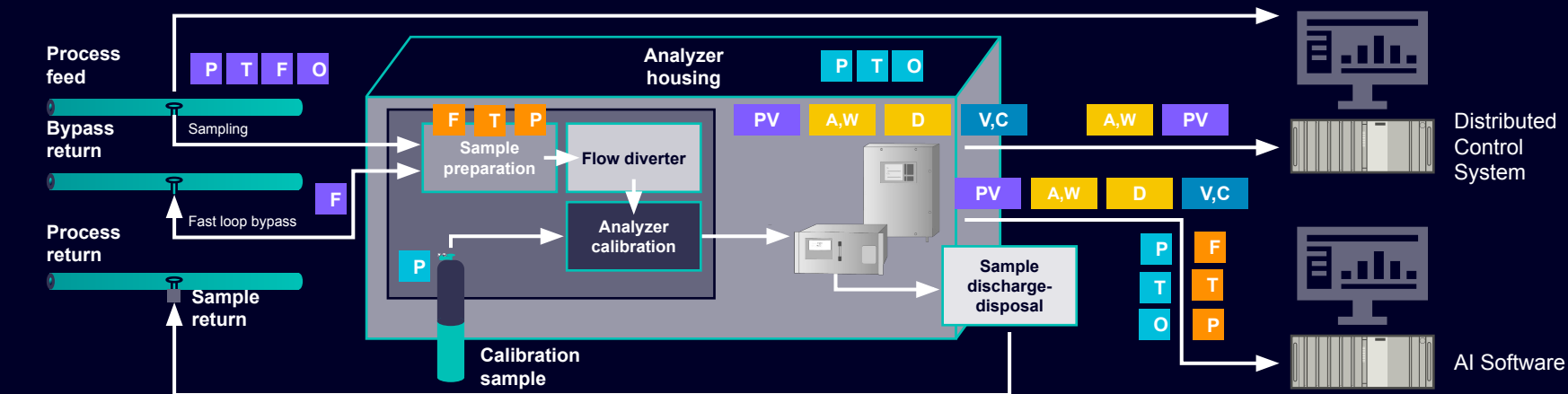
Reliable analyzers

Accurate, stable & available information

Optimized process

Profitable process

Today: Much information available from process analyzer is not used, but is useful to enable predictive maintenance with better analytical performance and less maintenance



Analyzer Information

PV Process value

A,W Alarms, warnings

D Diagnostic parameter (device health)

V,C Validation/calibration data

Sample Preparation

F Flow **P** Pressure **T** Temperature

Shelter

P T O Pressure, temperature, other signals

P Reference gas cylinder pressure

Process

P T F O Other process information such as temperature, pressure, flow, etc.

We can use all these data to predict maintenance needs upfront

What are typical issues of gas analyzer in the field?



- Frequent Validation and Calibration required
- Sample Handling Challenges
- Issues in Measurement Path from Sensor to Detector

leads to **incorrect measurement**

- failure of components within analyzer

leads to **unavailable measurement** due to downtime

We need to solve these issues using digitalization and Artificial Intelligence

What is Artificial Intelligence?



What do we want to achieve?

“Artificial Intelligence is the theory and development of computer systems capable of learning from historical data and performing tasks that historically required human intelligence such as recognizing speech, making decisions, and identifying patterns.”

- **Continuous, 24/7 and automatized analysis of process analyzers (instead of 1x time per week)**
- **Early identification of possible issues to inform technicians as soon as possible**
- **Provide technicians live overview about health status of process analyzers and provide helpful tools**



How can AI make a difference?

Detect potential
problems as early as
possible

Continuously validate
the measurement

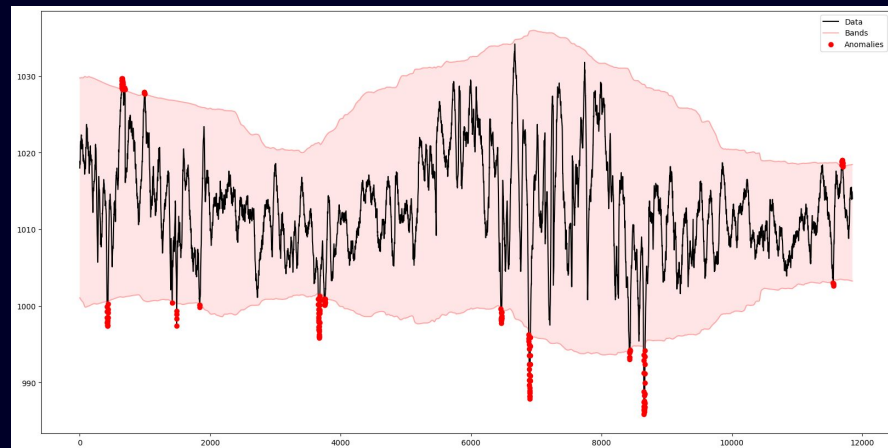
Provide replacement
value in case of
failure

Support technicians to solve problems fast

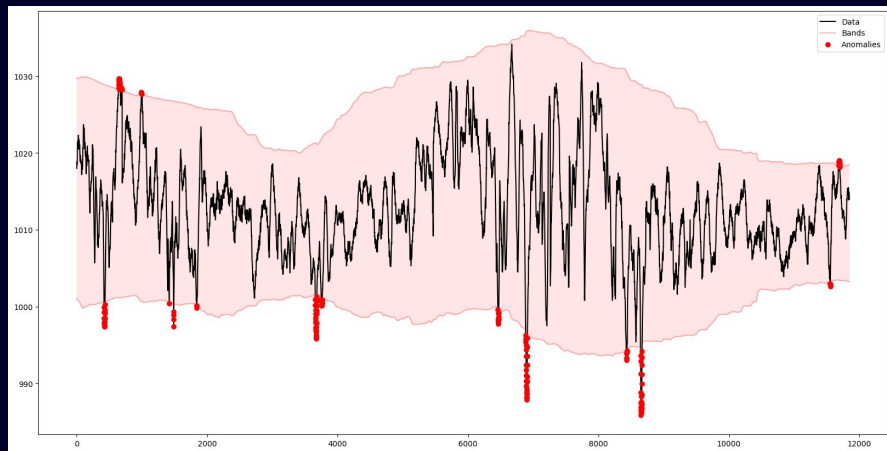
Detect potential problems as early as possible

What are AI systems capable of?

- AI learns past behavior of process analyzer and can conclude if live behavior is similar or not
- AI systems have the ability to improve itself over time by interaction with user (feedback from users or changes in overall process)
- Software informs technicians about potential issues which needs to be investigated further



Outlier Detection



Outlier Detection

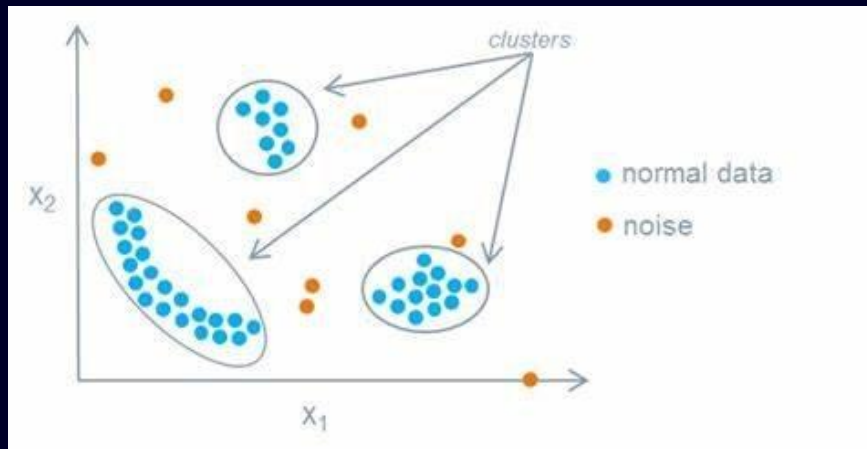
Scenario: Recognition of points outside acceptable operating range

Use Case:

- Measurement Value Outlier (further root cause analysis required if process or analyzer related)
- Sample Preparation outside of specification
- Problem within analyzer e.g. electronic board might have a problem

Value: Easy analysis without configuration of many limits which might be often not known (e.g. what is acceptable operating range of electronic board?)

Anomaly detection



Evaluation of behavior of analyzer, even if it is within defined operating levels

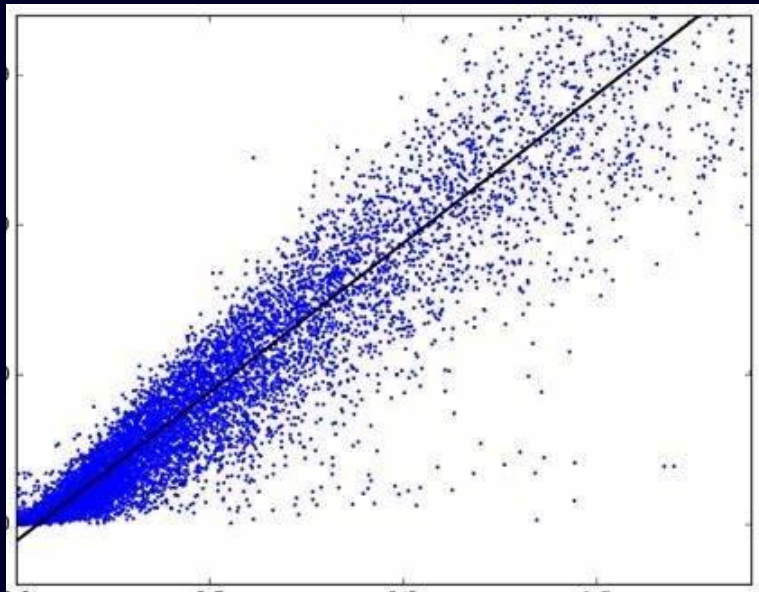
Scenario: Analyzed data is within limits, but strange pattern starts e.g. increasing fluctuation

Use Case:

- Typical behavior of components which soon will failure like electronics where smaller, spikes starts to occurs

Value: Early recognition if any strange behavior starts which give technicians advantage to react as soon as possible – even if device still operates.

Regression analysis



Regression Analysis to calculate possible impact of trend

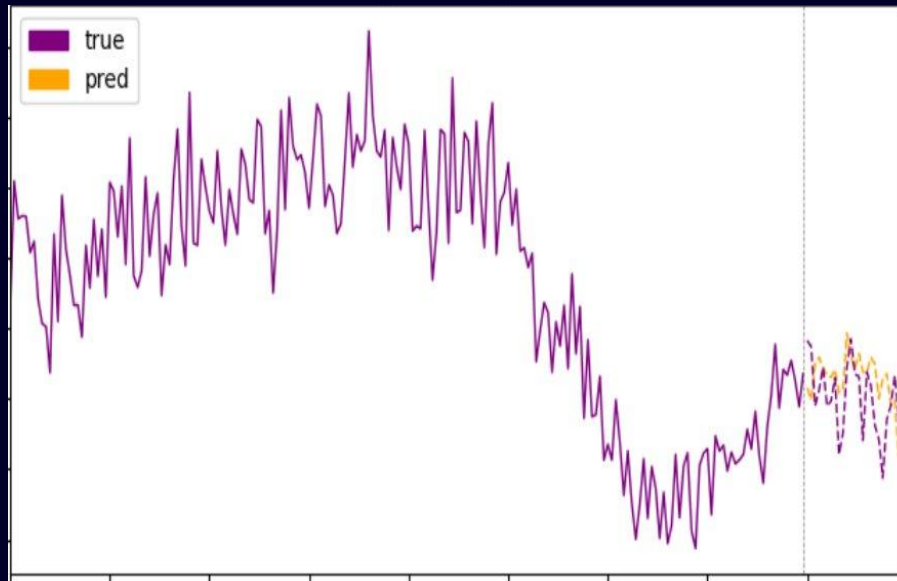
Scenario: Drift Recognition - What will happen if drift continues?

Use Case:

- Issues in Sample Preparation e.g. small pressure drop over time
- Sensitivity loss of sensor or detector due to aging effects (decrease over longer time)

Value: Calculation how much time left until analyzer system will operate outside of acceptable operating range and initiate proactive maintenance

Predict values in future



Prediction of values which is used as real-time check if analyzer health status is fine

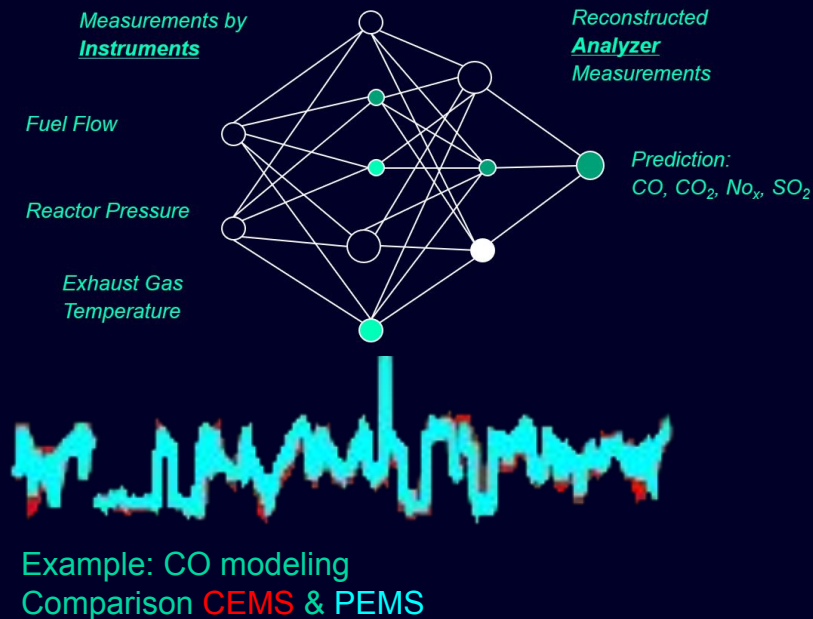
Scenario: Near-time future prediction of value for health status analysis or as back-up measurement

Use Case:

- Bridge over downtime of some hours by replacing substitution values to process control (with uncertainty)
- Use prediction as reference to check if there is any unusual behavior

Value: Live-check of analyzer + still deliver measurement results despite having downtime

Predict values in future – with even better performance



Use AI to provide measurement results for components which would require gas analyzer systems

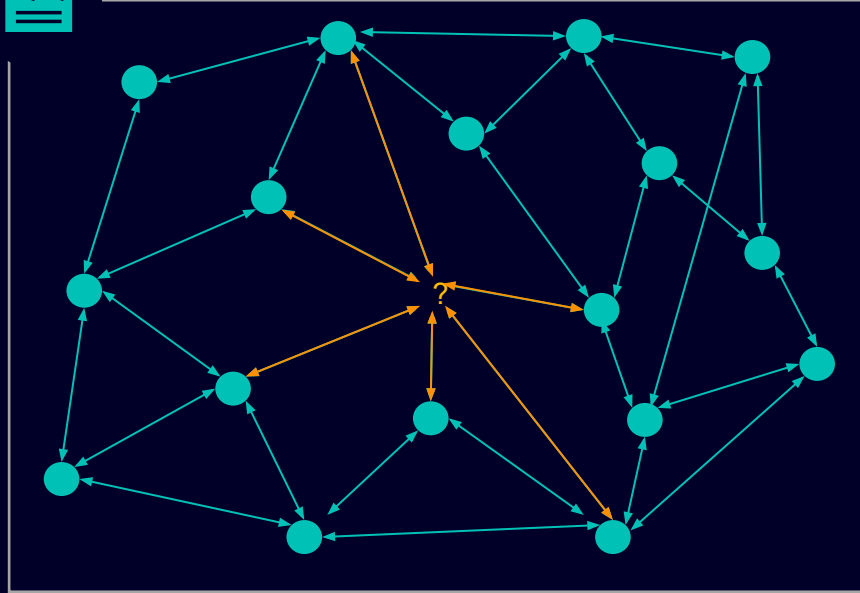
Scenario: Analyzer in downtime but process requires provision of measurement value

Use Case:

- Environmental authorities requires continuous emission measurement with high availability

Value: Provide back-up measurement with high accuracy despite downtime of gas analyzer system

Combining the data from sensor networks to enable virtual measurement



Process control requires specific measurement. Today's approach: New sensor accepting complexity and costs



Plant has already many sensors installed



Between them, there is a correlation defined by the process



Idea: Can we use these information to predict the target component without an additional sensor

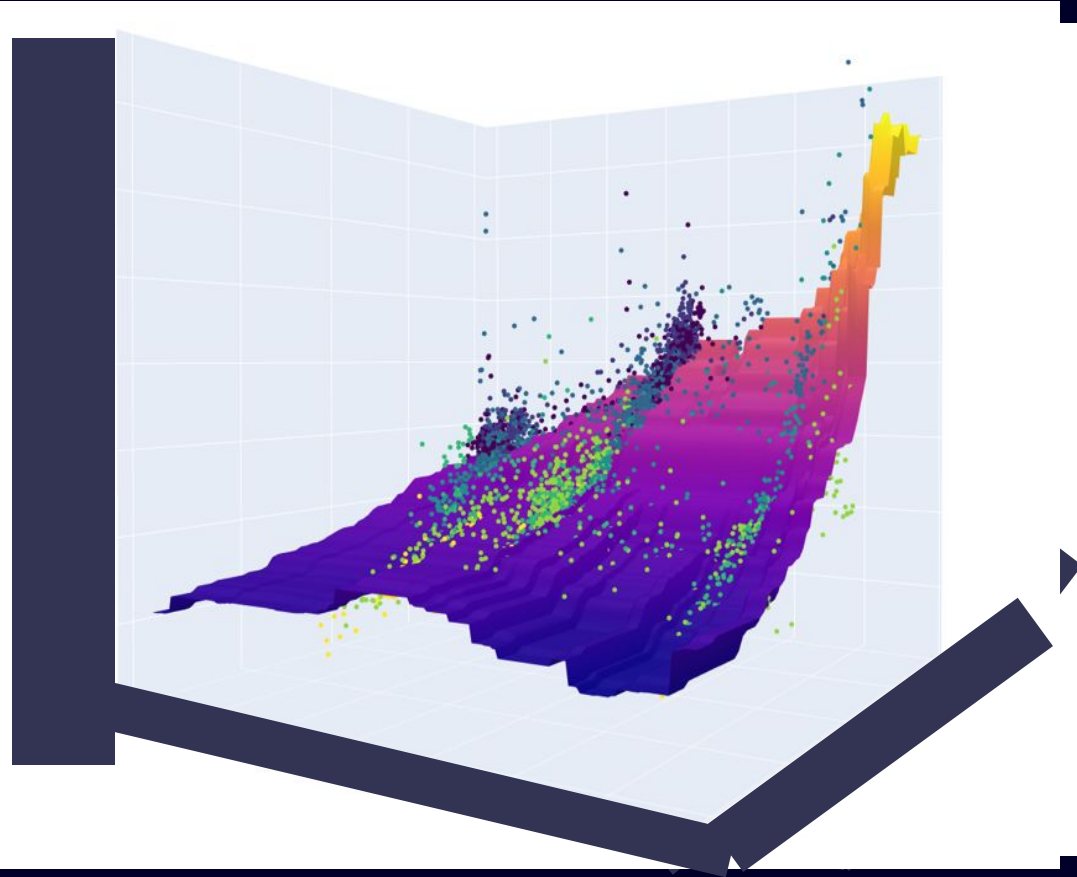


It's possible using a **soft sensor** based on process model

Use information from existing instruments in process to calculate gas composition

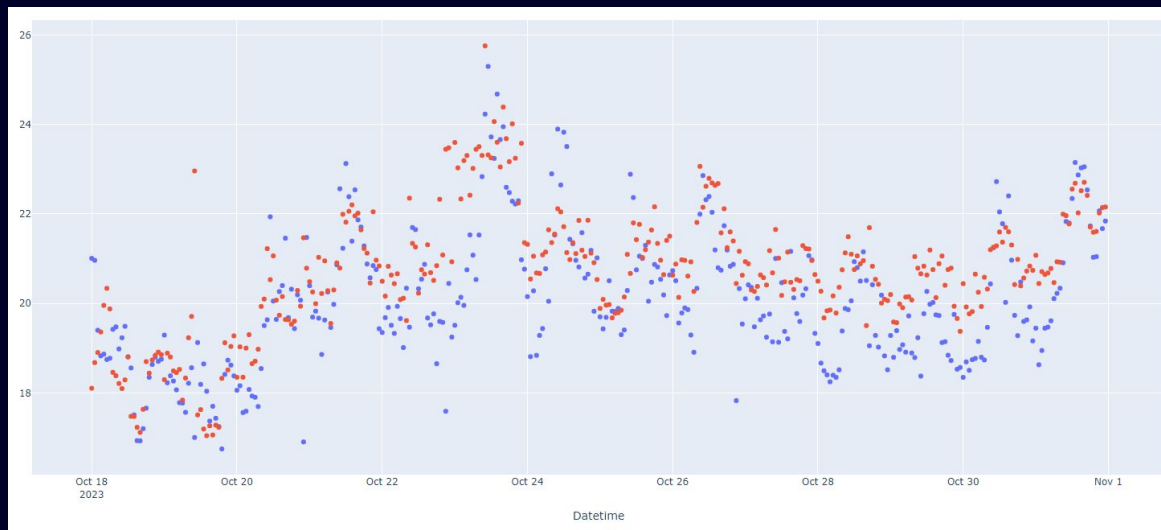
How does it work?

- AI is analyzing provided historical process data and is defining which parameters influence target values (in this case: emission values)
- Typical parameters are fuel flow rate, stack pressure and temperature as well as environmental conditions like humidity
- AI develops mathematical model based on historical data including input parameters and analyzer results and finetunes the model stepwise to come as close as possible to real values



Exemplary performance for NOx prediction in combustion process within refinery

Provision of
substitution values
is possible with high
accuracy up to 99%



Example shows relative accuracy of **soft sensor** for emission monitoring, in this case NOx in refinery in ASEAN. Accuracy deviation to **reference measurement** of below $\pm 3\%$, environmental specification required max $\pm 20\%$ deviation over test period.

More and more countries approve the usage of this technology as approved back-up or replacement of CEMS systems

Performance Specification 16

3/29/2023

While we have taken steps to ensure the accuracy of this Internet version of the document, it is not the official version. To see a complete version including any recent edits, visit: <https://www.ecfr.gov/cgi-bin/ECFR?page=browse> and search under Title 40, Protection of Environment.

PERFORMANCE SPECIFICATION 16—SPECIFICATIONS AND TEST PROCEDURES FOR PREDICTIVE EMISSION MONITORING SYSTEMS IN STATIONARY SOURCES

1.0 Scope and Application

1.1 Does this performance specification apply to me? If you, the source owner or operator, intend to use (with any necessary approvals) a predictive emission monitoring system (PEMS) to show compliance with your emission limitation under this part or 40 CFR part 61 or 63, you must use the procedures in this performance specification (PS) to determine whether your PEMS is acceptable for use in demonstrating compliance with applicable requirements. Use these procedures to certify your PEMS after initial installation and periodically thereafter to ensure the PEMS is operating properly. If your PEMS contains a diluent (O2 or CO2) measuring component, the diluent component must be tested as well. These specifications apply to PEMS that are installed under 40 CFR parts 60, 61, and 63 after April 24, 2009

EPA-PS16 influenced regulations:
US, Canada, Middle East, ASEAN (Malaysia, Thailand)

VEREIN DEUTSCHER INGENIEURE	Emissionen aus stationären Quellen Systeme zur Bestimmung von Emissionen mittels kontinuierlich überwachter Prozessparameter (PEMS) Grundlegende Aspekte, Eignungsprüfung, Zertifizierung und Anwendung	VDI-EE 3952
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Stationary source emissions – Predictive emission monitoring systems (PEMS) – Basic aspects, performance testing, certification, and application

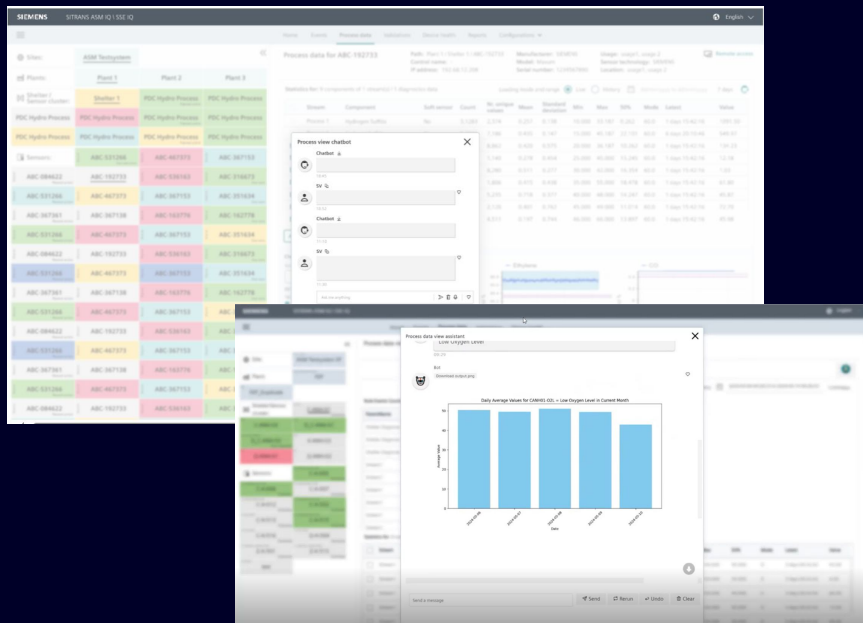
VDI-EXPERTENEMPFEHLUNG

Inhalt	Seite	Inhalt	Seite
Vorbemerkung	2	8.6 Funktionsprüfungen	26
Einleitung	2	8.7 Berichterstattung	30
1 Anwendungsbereich	2	9 Dokumentation	30
2 Normative Verweise	3	Anhang A Statistische Kenngrößen	31
3 Begriffe	3	A1 Bestimmtheitsmaß	31
4 Formelzeichen und Abkürzungen	4	A2 Relative Genauigkeit	31
5 Grundlegende Aspekte	5	Anhang B Mindestanforderungen an Systeme zur Bestimmung von Emissionen mittels kontinuierlich überwachter	
5.1 Allgemeines	5		

VDI-EE 3952:
Germany, expected to be taken over in Europe



Next level of maintenance using Generative AI



Future of analyzer maintenance using Chatbot based on Generative AI

Scenario: Technicians want to find out how to solve a specific occurred problem

Use Case:

- Provide recommendations automatically out of manuals without search
- Show similar cases in the past with same or comparable analyzers

Value: Make it easier to maintain process analyzer by making expert knowledge available for everyone

Conclusion



- ✓ Powerful Artificial Intelligence can be used to detect issues with analyzer as soon as possible
- ✓ AI can be used to simplify maintenance e.g. by providing historical recommendations using GenAI
- ✓ Process Analyzers can be backed-up by solutions like PEMS to provide values despite downtime

Digitalization combined with reliable gas analyzer systems are key for next level of automation and process optimization